COMP 3920 Database Systems 1

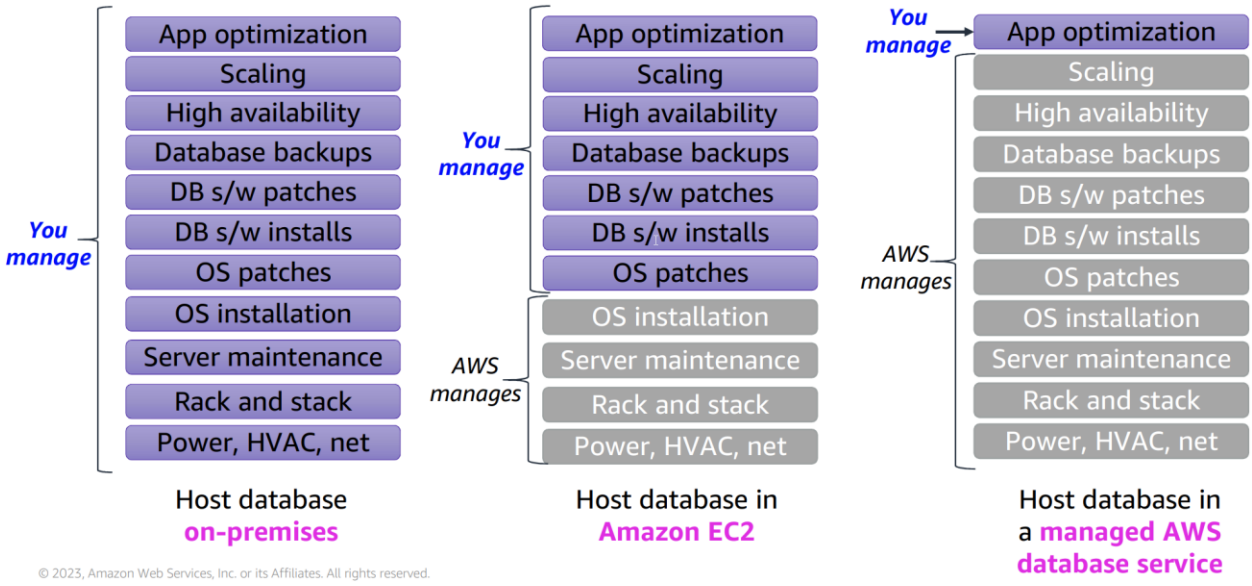
Amazon Web Services (AWS) Relational Database Service (RDS) Part 1

Introduction:

The goal of this document is to serve as an introduction to using and managing Amazon's Relational Database Services (AWS RDS).

Amazon's Relational Database Services (RDS), which are part of Amazon's ever growing sets of cloud services called Amazon Web Services (AWS), allows for creating relational databases, in many popular database systems such as: PostgreSQL, Oracle, MS SQL, MariaDB and MySQL. The difference however, between purchasing your own server and installing MySQL or running an AWS Elastic Cloud Computing (EC2) instance with MySQL installed, is that RDS manages the hardware, database setup, updates and backups. RDS can even help with creating and managing MySQL replication.

As shown the diagram below, Amazon RDs handles much of the server management tasks (such as hardware, power, OS and database installation, patches and backups). This gives you more time to focus on your application design, development and optimization.



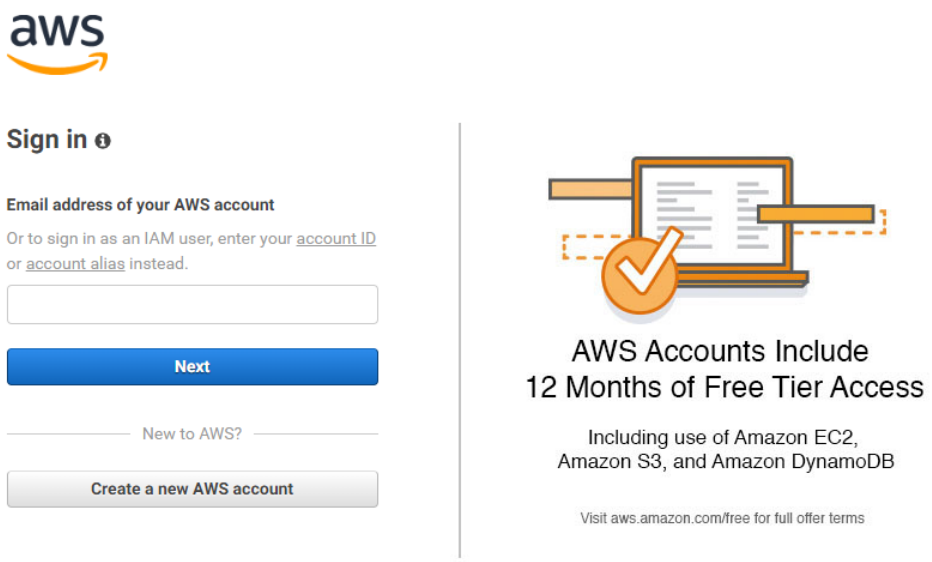
In this lab we will:

- Create an AWS RDS instance.
- Connect to and run SQL Statements on our RDS Instance.
- Backup our RDS instance (create an RDS Snapshot).
- Restore from an RDS Snaphot.

Step 1:
Sign in to AWS.

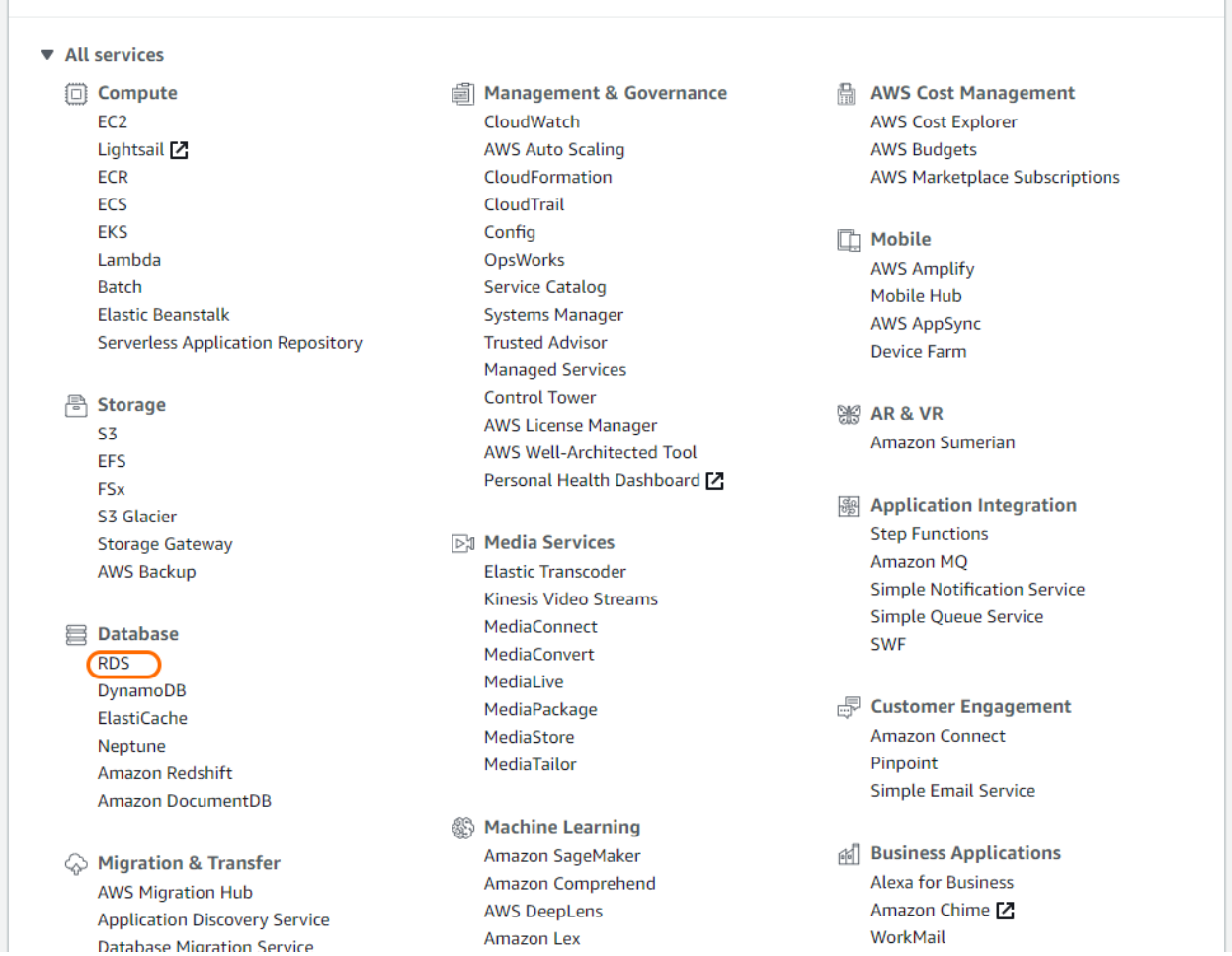
Go to: https://console.aws.amazon.com

Enter your credentials to sign in.

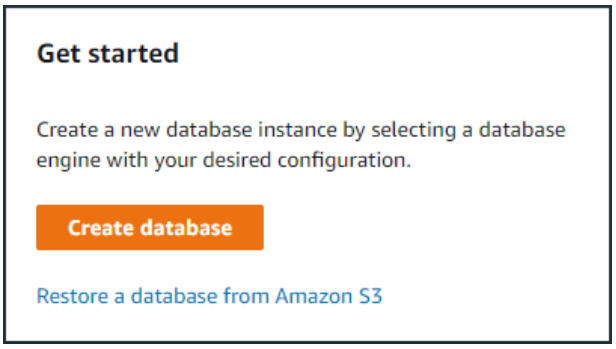


Step 2:
Create a MySQL RDS Database Instance.

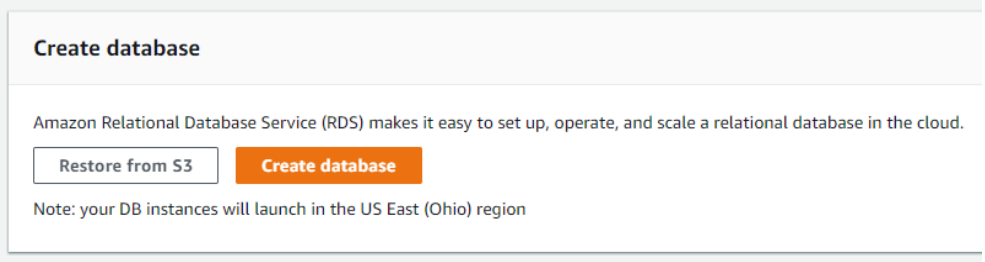
Find the "RDS" Service under the Database Section.



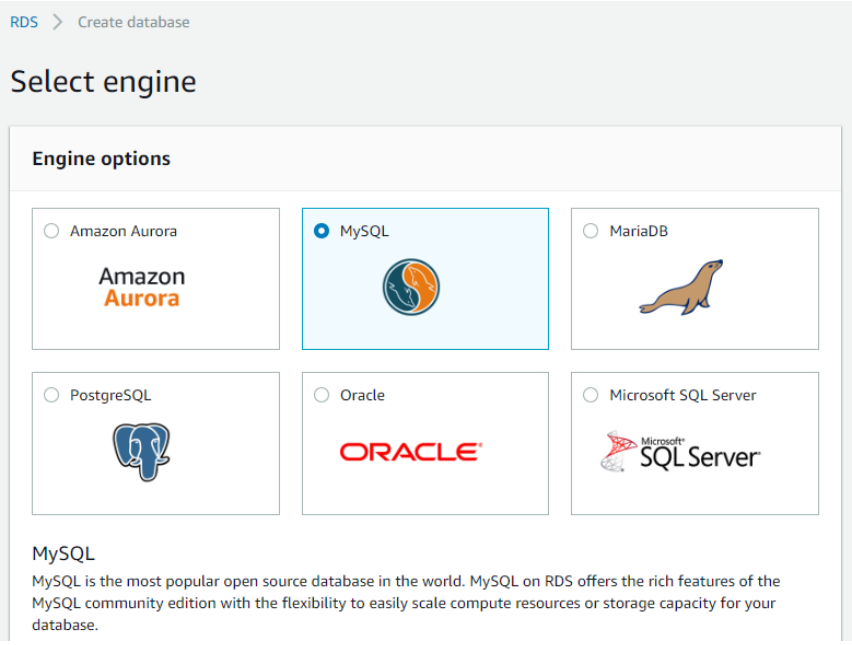
Create a new RDS Instance, either in the Get started section by clicking "Create database"…



… or by clicking "Create database" in the Create database section.

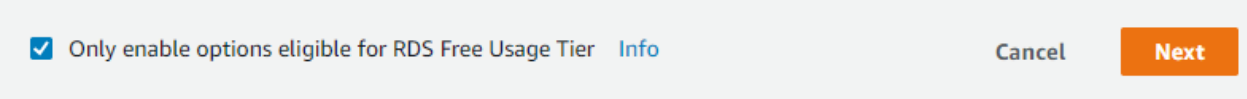


Select MySQL as your Database Engine.



IMPORTANT: To restrict the instance to only use the AWS Free Tier (so you don't have to pay for it) check off "Only enable options eligible for RDS Free Usage Tier".

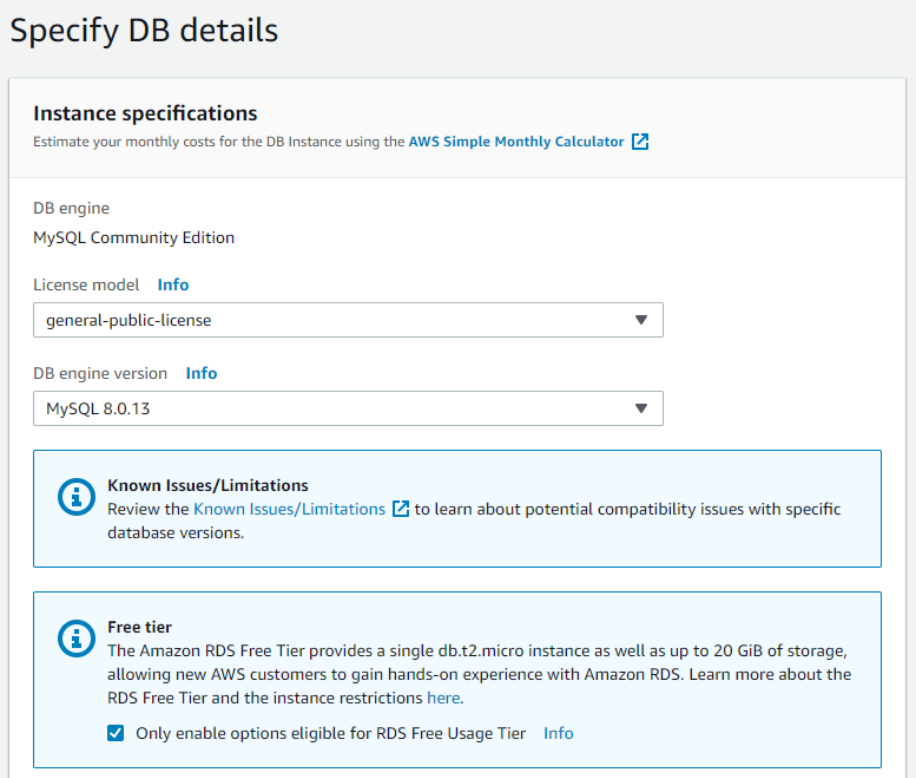
Save yourself some money!



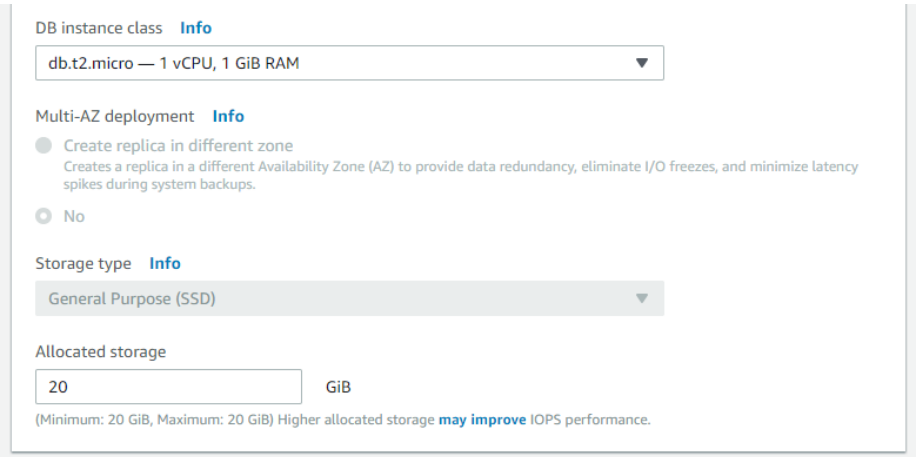
Click Next.

Instance Specifications:

Select the latest version of MySQL 8.0 (in my case MySQL 8.0.13).
Leave the licensing model as is, and ensure the Free Tier is still checked.



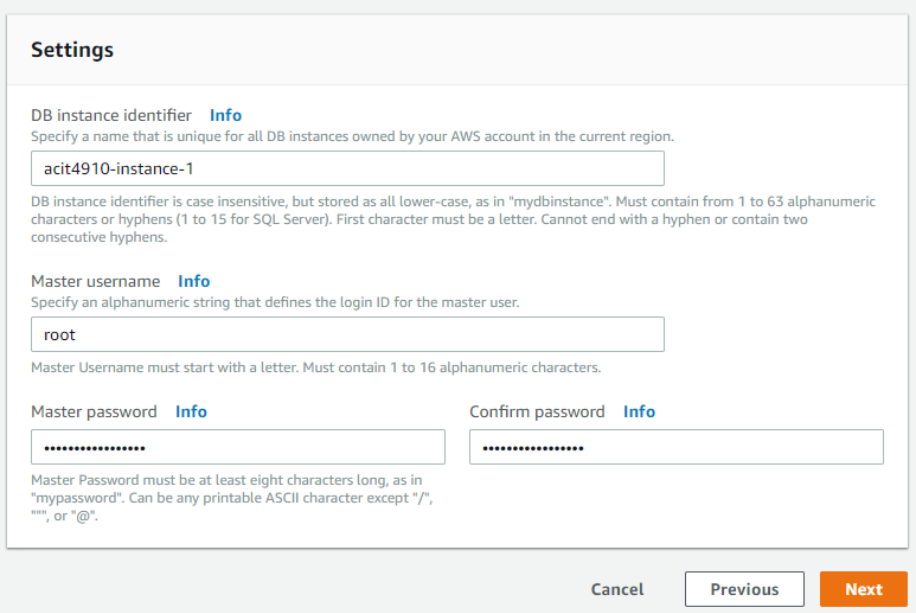
Your DB instance class will be limited to the smallest (free tier) instance type, most likely db.t2.micro.
This is ok.
20 GiB should be plenty for our database to start.



Settings:

Call your instance: **comp3920-instance-1**.
Create a Master username: **masteruser**
Create a new Password: **<your master password>**

Click Next.

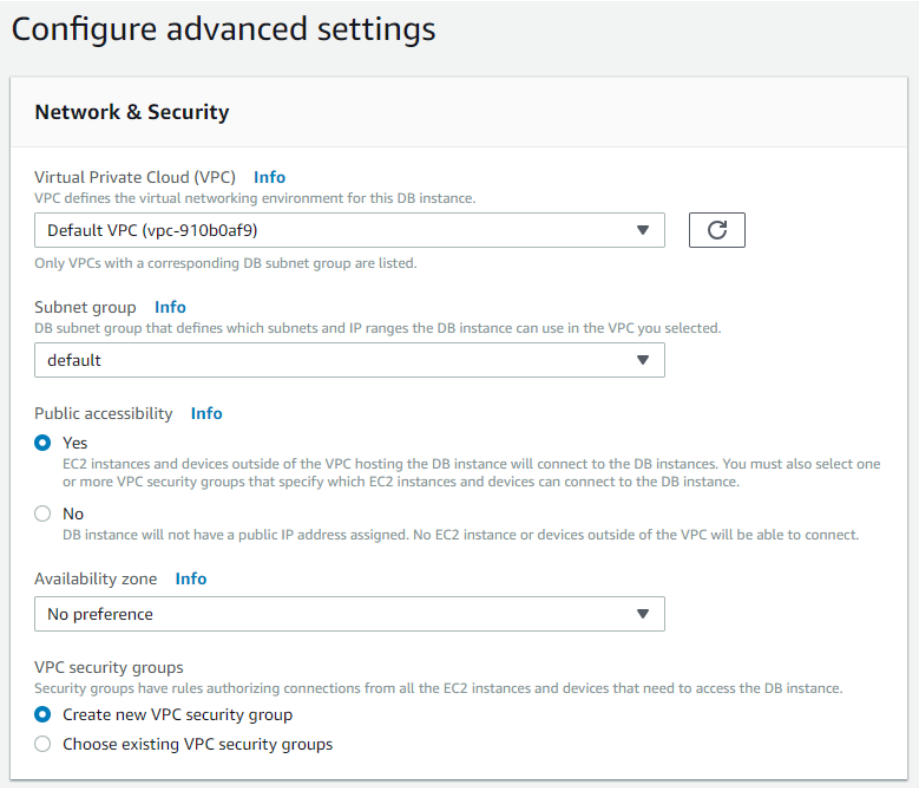


Network & Security:

On the next page of advanced settings, you can configure which network (or virtual network) your new database instance will be on. This will be very helpful if you needed to connect your instance to another instance for copying data, replication etc. You would also want to take note of the Virtual Private Cloud (VPC) if you wanted to connect a webserver such as an AWS EC2 Instance to your database. All of your instances that need to communicated with each other should be on the same VPC.

To connect an AWS EC2 Instance to an AWS RDS Instance do they need to be on the same Virtual Private Cloud?

Since this is our first instance it doesn't matter which VPC we use.
Create as Publicly accessible.
Create new VPC security group.



Connectivity:

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- (●) Don't connect to an EC2 compute resource
  Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- ( ) Connect to an EC2 compute resource
  Set up a connection to an EC2 compute resource for this database.

We currently don't need an EC2 compute resource to connect to our database. Leave this as "Don't connect". We could use this option to create an additional EC2 instance with all the same VPC and related network settings (such as security groups, subnets, etc.) to be able to connect to the database. Additional information: https://aws.amazon.com/about-aws/whats-new/2022/08/amazon-rds-setting-up-connectivity-rds-database-ec2-compute-instance-1-click/
and
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_CreateDBInstance.html#USER_CreateDBInstance.Prerequisites.VPC.Automatic

Database options:
We don't want to create a database just yet, so leave the Database name empty.
Leave the port as **3306**.

Database options

Database name Info
dbname
Note: if no database name is specified then no initial MySQL database will be created on the DB Instance.

Port Info
TCP/IP port the DB instance will use for application connections.
3306

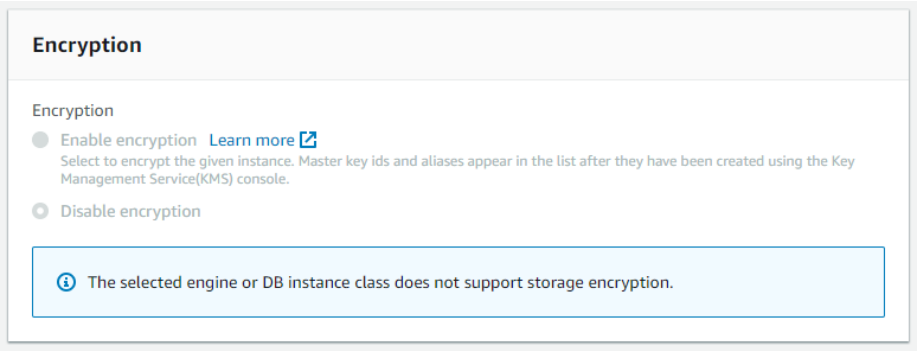
DB parameter group Info
default.mysql5.6

Option group Info
default:mysql-5-6

IAM DB authentication Info
- ( ) Enable IAM DB authentication
  Manage your database user credentials through AWS IAM users and roles.
- (●) Disable

Encryption:

Encryption is not supported. Leave this section as is.



Backup:

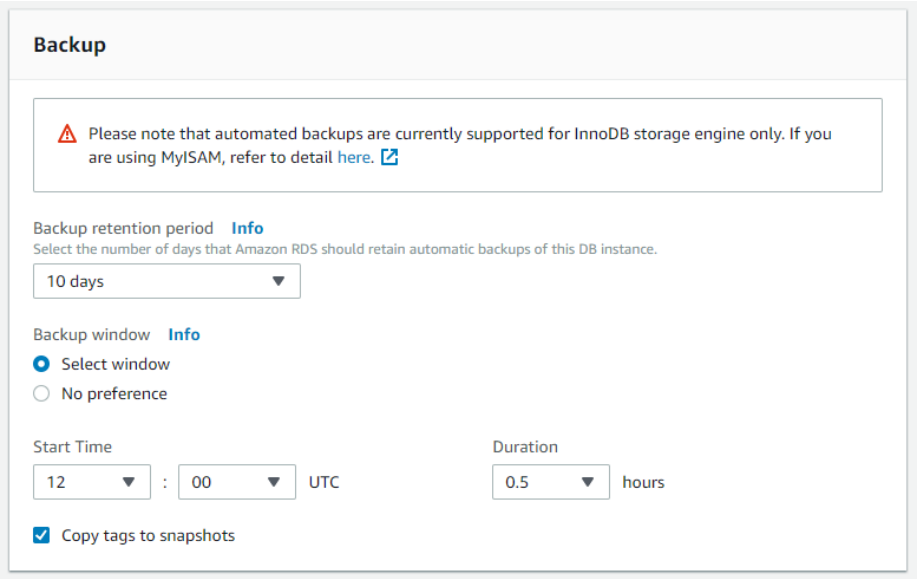
Keep your backups for: **10 days**.

Select a maintenance window for starting at **4:00 AM Pacific Time** [UTC-8] (12:00 noon UTC), lasting for **30 minutes**.

Take note of the warning about InnoDB and MyISAM. Click on the link by "refer to detail here" to read more.

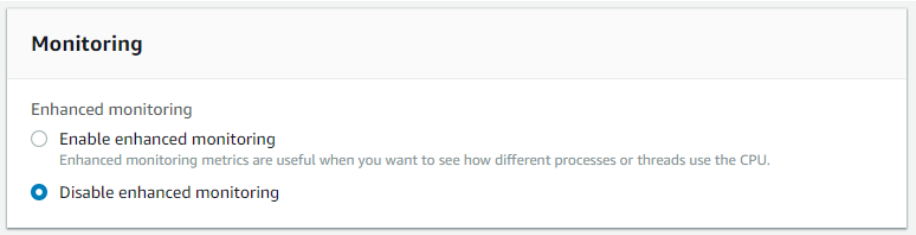
Does Amazon's RDS support MySQL's table storage engines MyISAM and InnoDB?
Are both MyISAM and InnoDB safe for backups and recovery?

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_MySQL.html#MySQL.Concepts.Storage



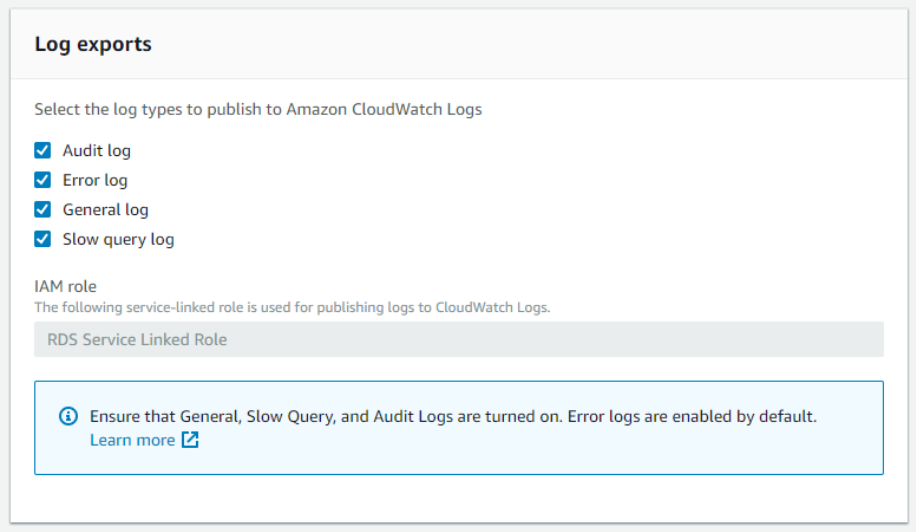
Monitoring:

Leave enhanced monitoring **disabled**.



Log Exports:

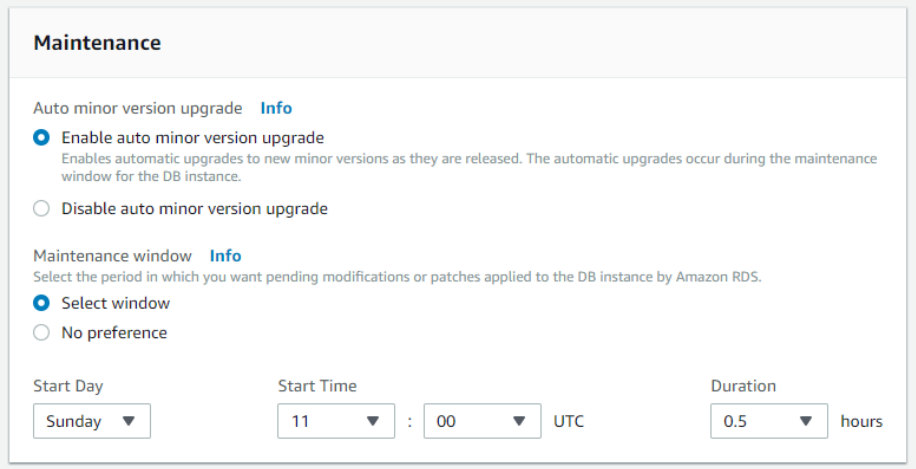
Select all logs: Error Log, General Log, Slow Query Log (and Audit Log if available).



Maintenance:

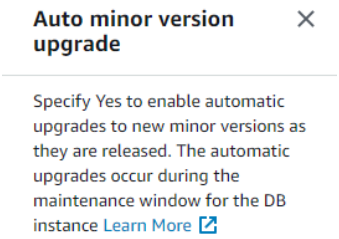
Enable minor version updates.

Select a maintenance window for starting **Sunday at 3:00 AM Pacific Time** [UTC-8] (11:00 UTC), lasting for **30 minutes**.



With Amazon's RDS, minor version upgrades (if enabled) will automatically update MySQL to the latest *minor version*. This can help reduce security risks by patching known security vulnerabilities.

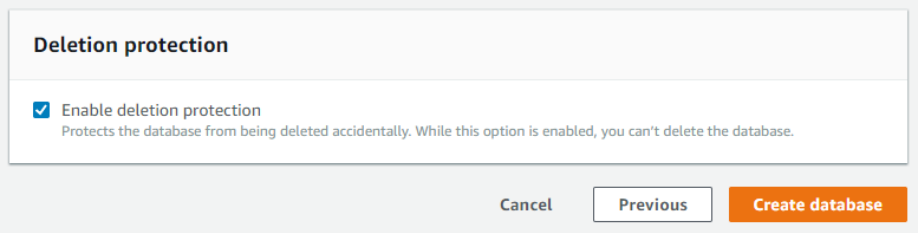
Take note about the information on minor version *upgrades.* Click on the link "info", then "Learn More" on the right to read more.



What version(s) does Amazon RDS support for upgrades?
What is a minor version?

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_MySQL.html#MySQL.Concepts.VersionMgmt

<u>Deletion protection:</u>
Keep deletion protection turned on.



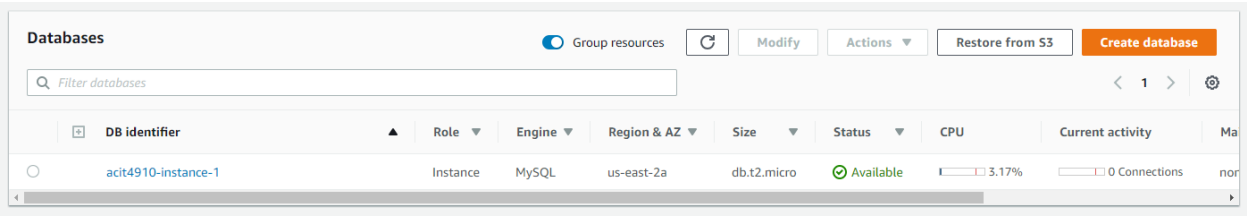
Before you create your new database. We are going to time how long it takes to create our new database instance.

Open up a stopwatch (within Windows you can use the Alarms and Clock App):

Time how long it takes from when you click "Create Database" to when the instance has been created.

Click "Create Database".

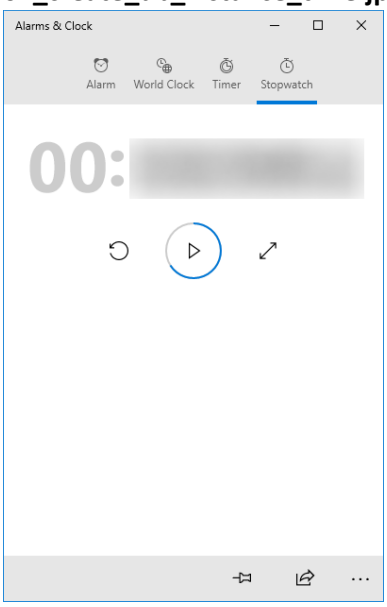
IMPORTANT: Go to this screen (shown below) and wait for Status: **Available**, then stop your timer. This screen, updates better and faster than the other screens.



This may take anywhere from 5 to 15 minutes or more. But it is still faster than buying your own in-house server, downloading MySQL from their website, installing and configuring MySQL, AMIRITE?

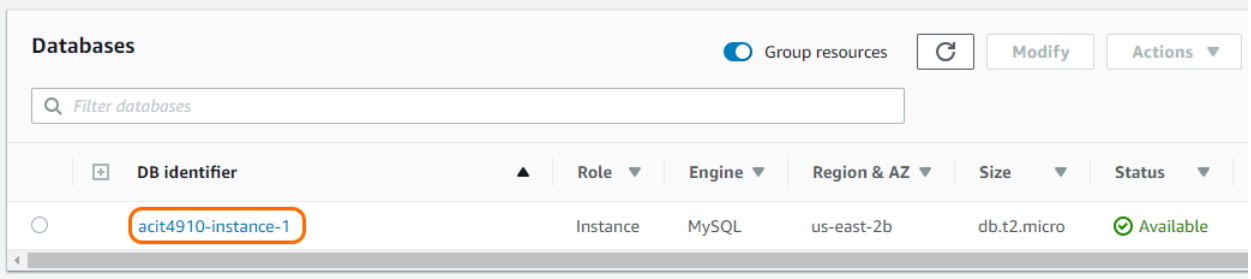
Submit your Screenshot:
How long did your database instance take to create?
Submit your stopwatch.

01_create_db_instance_time.jpg Submit

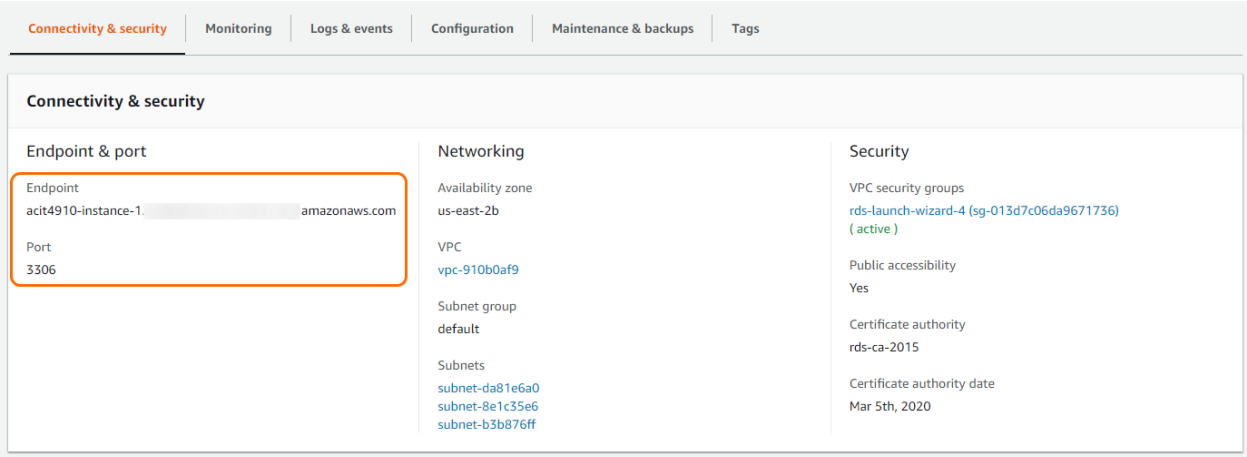


Step 3:
Connect to your new RDS instance.

Click on your RDS instance name to open the instance details.



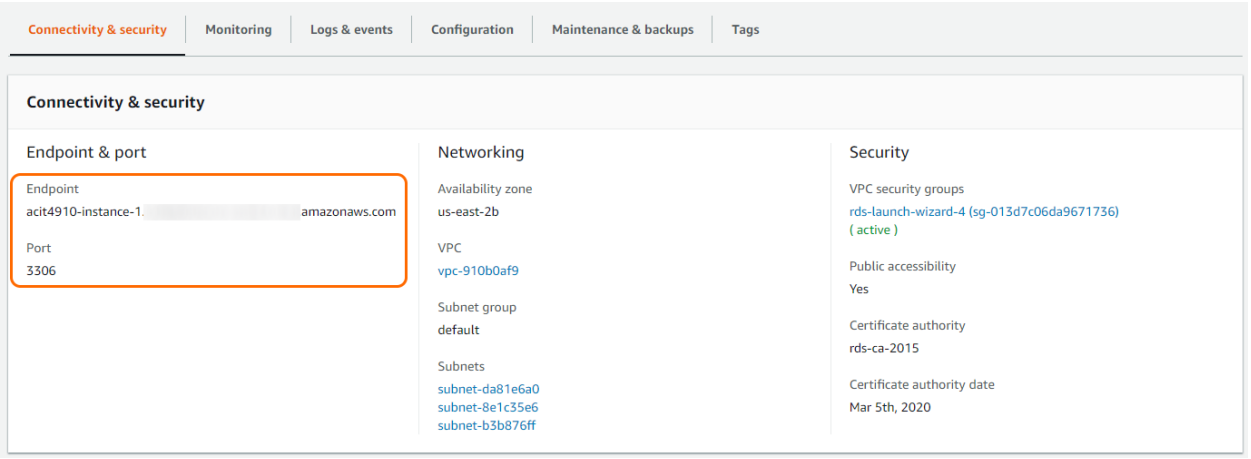
Locate your endpoint and port.



Submit your Screenshot:
Show the endpoint and port of your new instance

02_endpoint_and_port.jpg

Submit



From a computer (PC, VM or EC2 instance) that has the MySQL Client program, connect to the endpoint of your AWS RDS instance.

```
mysql -u masteruser -p -h <your RDS instance endpoint>
```

Show the current running version of MySQL:

```
SHOW variables LIKE 'version';
```

Submit your screenshot:
Your screenshot must show the connection command **and** the version information.

Submit

03_connection_successful.jpg

```
user@ubuntu-VirtualBox:~$ mysql -u masteruser -p -h acit4910-instance-1.[redacted].rds.amazonaws.com
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 135
Server version: 8.0.13 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show variables like 'version';
+---------------+--------+
| Variable_name | Value  |
+---------------+--------+
| version       | 8.[redacted]|
+---------------+--------+
1 row in set (0.09 sec)
```

Step 4:
Explore your brand-new database instance.

Show the databases that come pre-installed with your RDS instance.

```
SHOW DATABASES;
```

What databases do you have?

What users come installed on your database instance?

```
SELECT user, host, plugin FROM mysql.user;
```

Most brand new MySQL installations include the following users:

```
root@localhost
mysql.infoschema@localhost
mysql.session@localhost
mysql.sys@localhost
```

Are your RDS users different?

Step 5:
Create a simple database.

Create the `zoo` database.
```
CREATE DATABASE zoo;
```

Create a `mammal` table.
```
CREATE TABLE `zoo`.`mammal` (
  `mammal_id` INT NOT NULL AUTO_INCREMENT,
  `name` VARCHAR(45) NULL,
  PRIMARY KEY (`mammal_id`));
```

Insert a **bat**, **rabbit** and **kangaroo** in the `mammal` table.

```
mysql> select * from zoo.mammal;
+-----------+----------+
| mammal_id | name     |
+-----------+----------+
|         1 | bat      |
|         2 | rabbit   |
|         3 | kangaroo |
+-----------+----------+
3 rows in set (0.12 sec)
```

Create a `reptile` table.

```
CREATE TABLE `zoo`.`reptile` (
  `reptile_id` INT NOT NULL AUTO_INCREMENT,
  `name` VARCHAR(45) NULL,
  PRIMARY KEY (`reptile_id`));
```

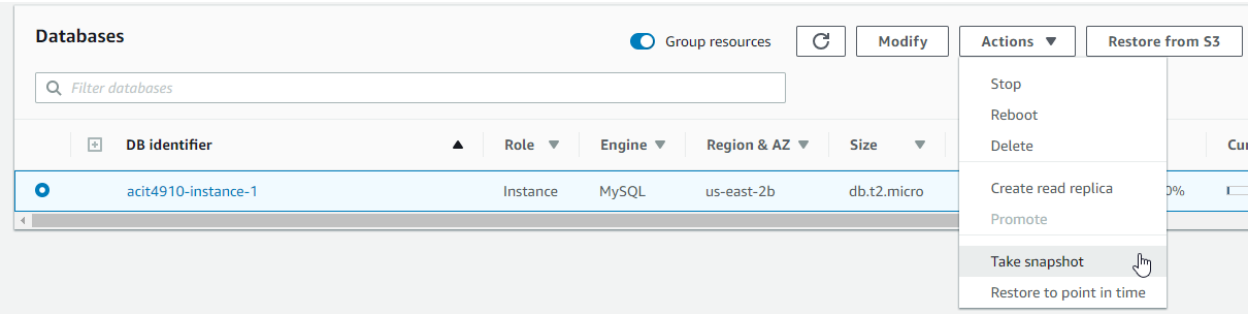
Insert a **snake**, **gecko** and **chameleon** in the `reptile` table.

```
mysql> select * from zoo.reptile;
+------------+-----------+
| reptile_id | name      |
+------------+-----------+
|          1 | snake     |
|          2 | gecko     |
|          3 | chameleon |
+------------+-----------+
3 rows in set (0.08 sec)
```

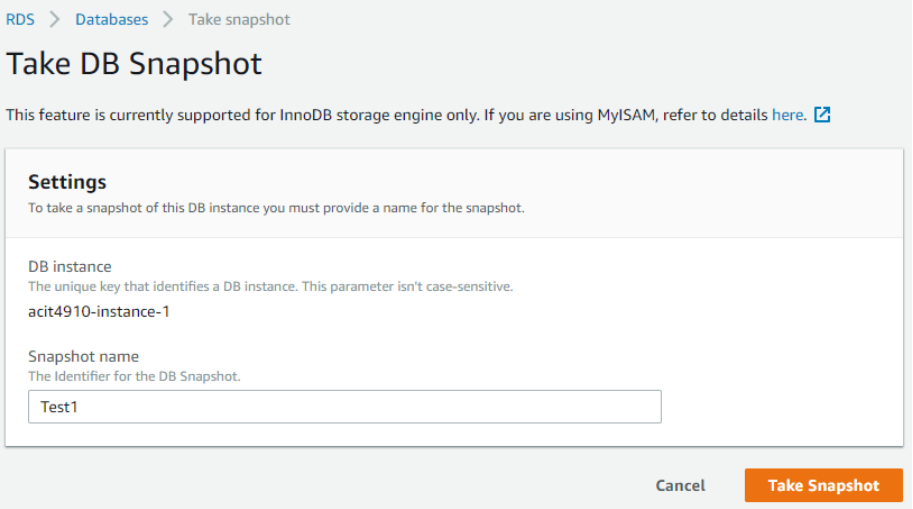
Step 6:
Backup your RDS database instance.

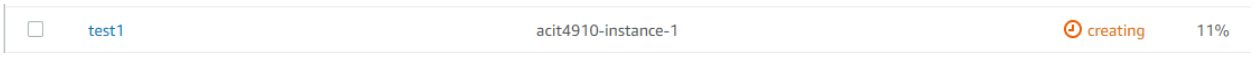
In Amazon's RDS interface, backups are called *snapshots*. Take a snapshot of your database.

Select your RDS instance by clicking the circle to the left of the instance name (comp3920-instance-1). Click on the "Actions" button, and click "Take snapshot".



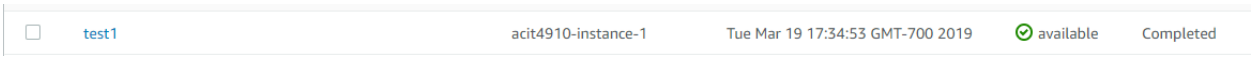
Name your Snapshot `test1`.





While Amazon takes the snapshot, create a new table `bird`.

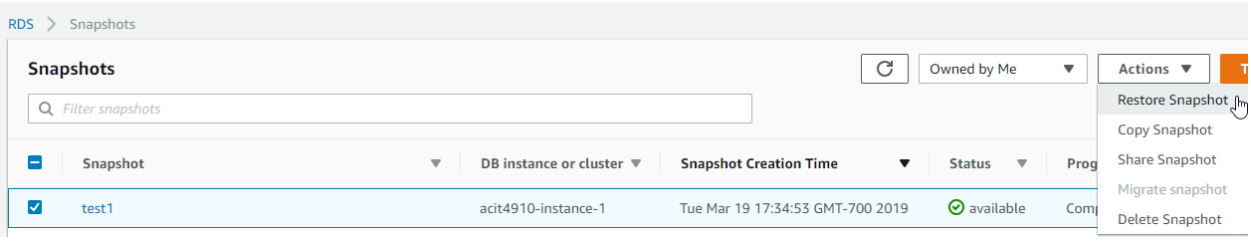
```
CREATE TABLE `zoo`.`bird` (
  `bird_id` INT NOT NULL AUTO_INCREMENT,
  `name` VARCHAR(45) NULL,
  PRIMARY KEY (`bird_id`));
```



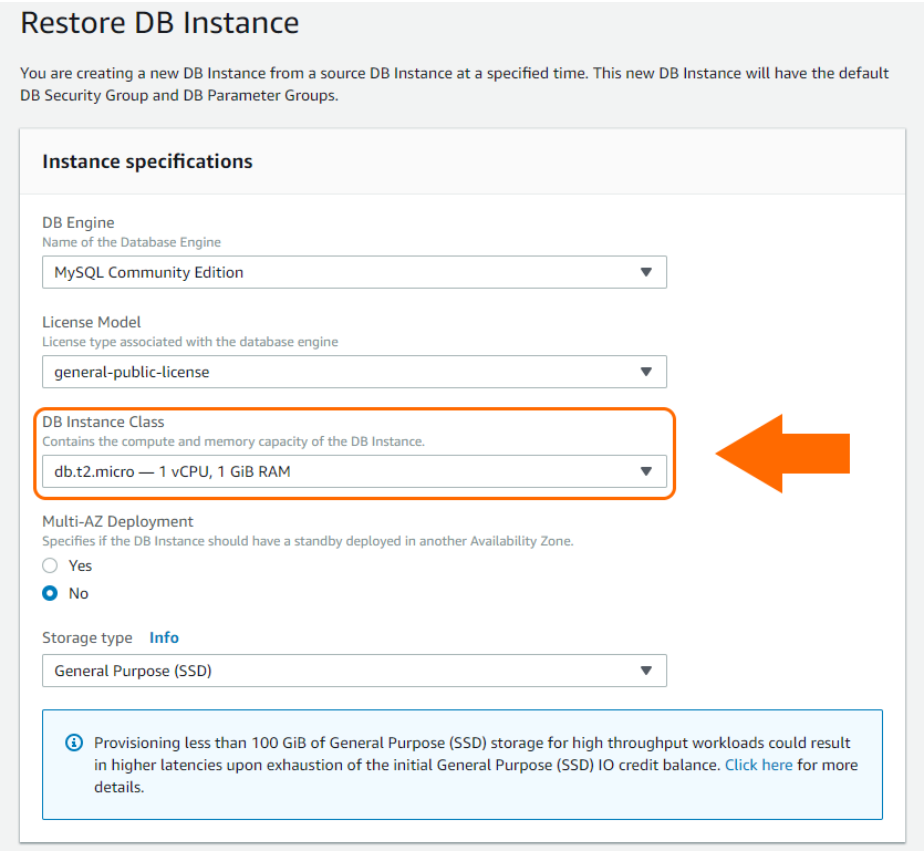
What happens when you try to create your new table bird while the snapshot is in progress? Does it succeed?

Step 7:
Restore from Snapshot.

Select your `test1` snapshot, click "Actions" and click "Restore Snapshot".

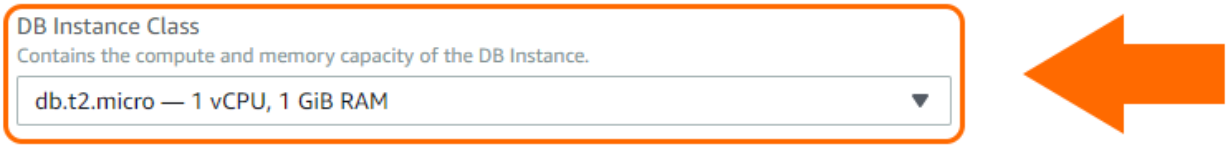


Many of these settings will be familiar to you because they are the same as when you create an instance from nothing. The reason for this is, in Amazon's RDS, when you restore from a snapshot, you create a brand-new instance. This is the ***only*** way to restore a database in RDS.

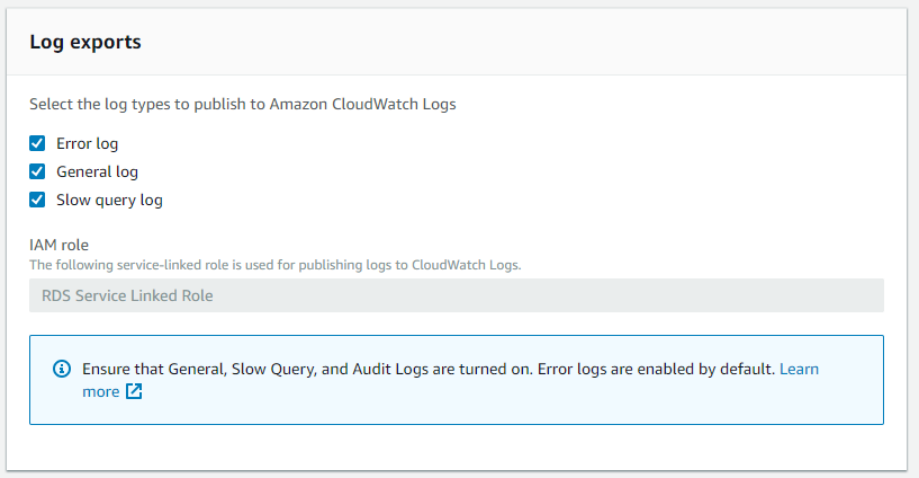


Many of these settings will be left as a default with the following exceptions:

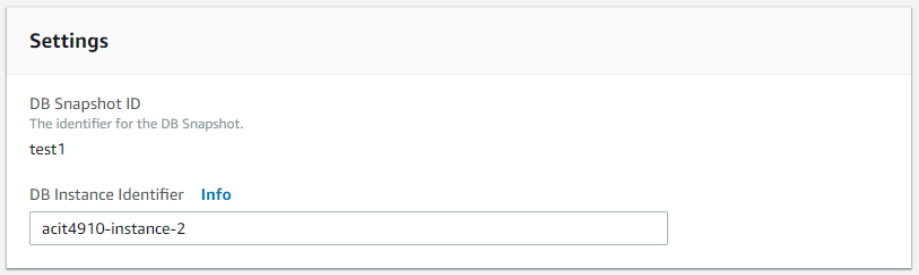
IMPORTANT: Set your DB Instance class to `db.t2.micro`. Otherwise you won't be on the Free Tier anymore.



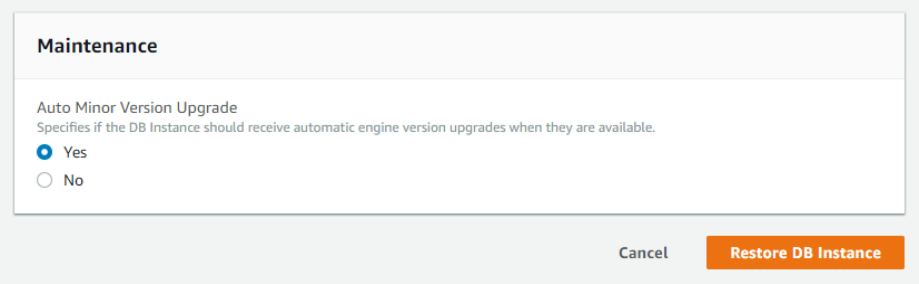
Enable all Logs: Error Log, General Log, Slow Query Log (and Audit Log, if available).



Name your instance `comp3920-instance-2`.



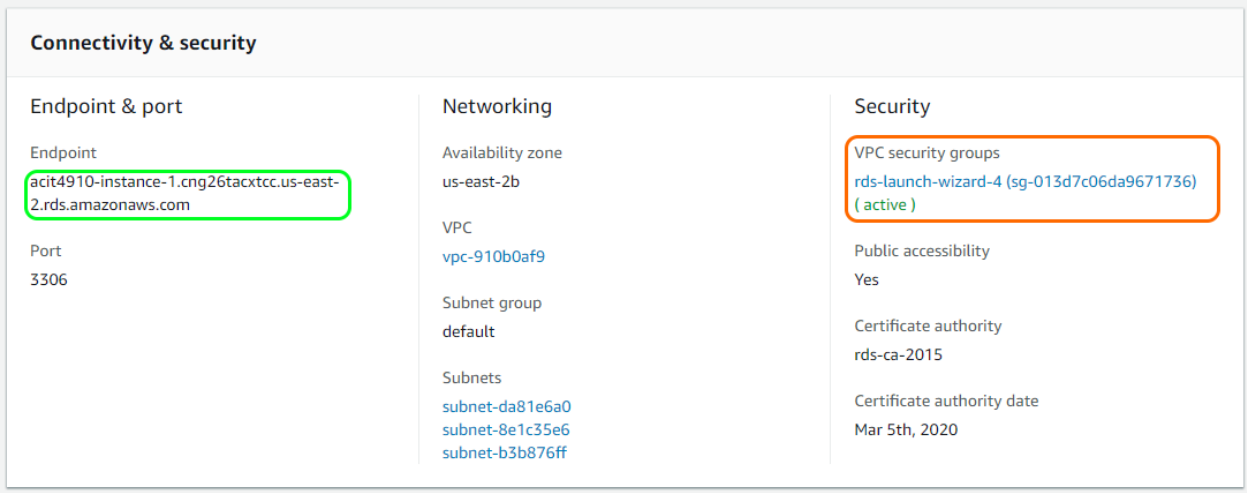
Click Restore DB Instance.



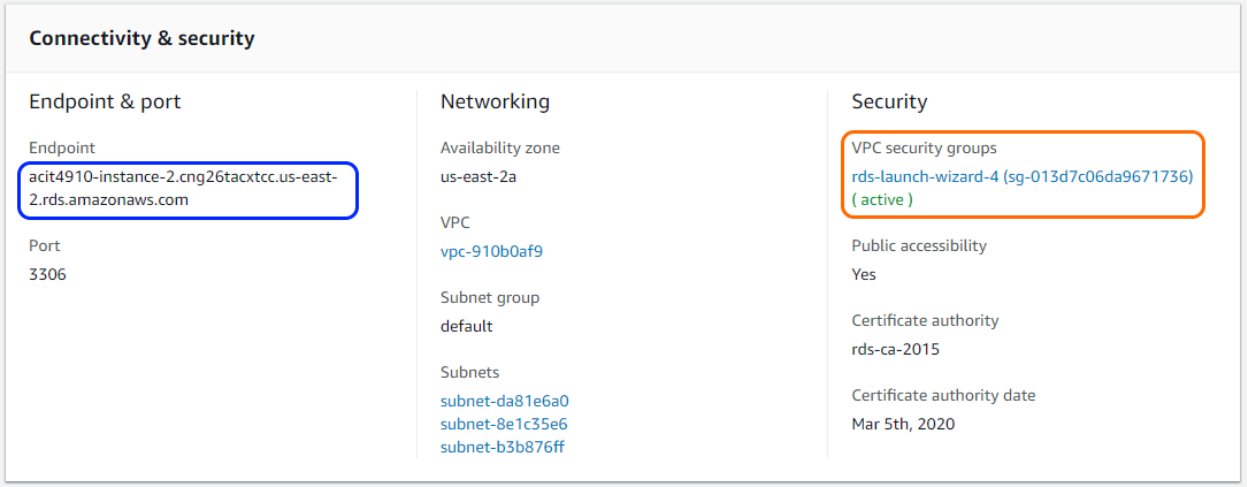
When restoring from a snapshot sometimes a new security group is created that doesn't allow incoming connections on port 3306 from your IP address.

Confirm that the security group for RDS `comp3920-instance-2` is the same as `comp-instance-1`.

Security Group for **comp3920-instance-1**.



Security Group for **comp3920-instance-2** (should match the security group for **comp3920-instance-1**).



Step 8:
Confirm that our restore from snapshot worked.

Login to RDS instance 2 (`comp3920-instance-2`).
Do you remember how to get the endpoint for an RDS instance?

List the databases on instance 2.

Submit your screenshot:
Your screenshot must show the connection command **and** the show databases information.

04_restore_successful.jpg

Submit

```
user@ubuntu-VirtualBox:~$ mysql -u masteruser -p -h acit4910-instance-2.[redacted].rds.amazonaws.com
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.13 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| zoo                |
+--------------------+
5 rows in set (0.09 sec)
```

Submit your screenshot:
Your screenshot must show the connection command **and** the show databases information.

05_consistent_snapshot.jpg

Submit

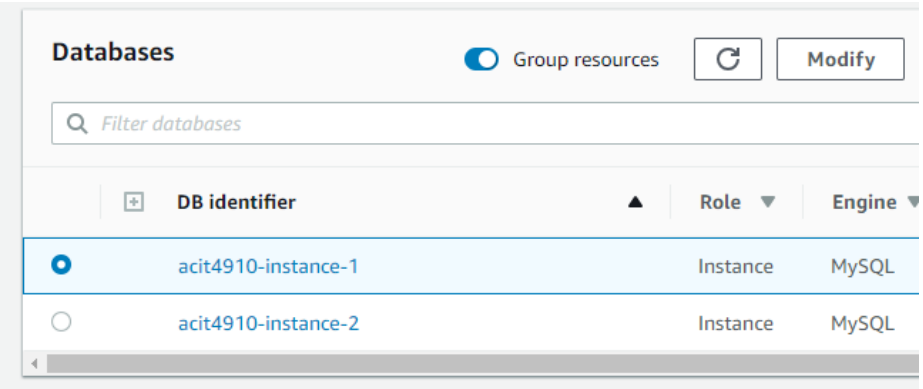
```
mysql> show tables from zoo;
+----------------+
| Tables_in_zoo |
+----------------+
[redacted]
+----------------+
 rows in set (0.10 sec)
```

Did our `bird` table, that was created *after* the snapshot started, but *before* the snapshot finished, get included in our snapshot?

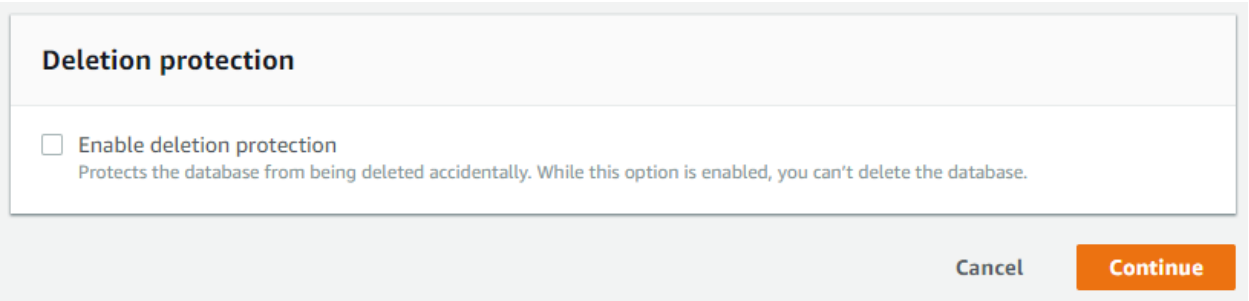
IMPORTANT: Clean up your AWS Instances when you are done the lab!

Even though we created our instances as db.t2.micro (Free Tier), we should clean up instances just in case. Also, since we don't need our snapshots, we should delete those as well. The storage that is required to keep our snapshots may end up costing us money as well.

Select `comp3920-instance-1` and click "Modify".

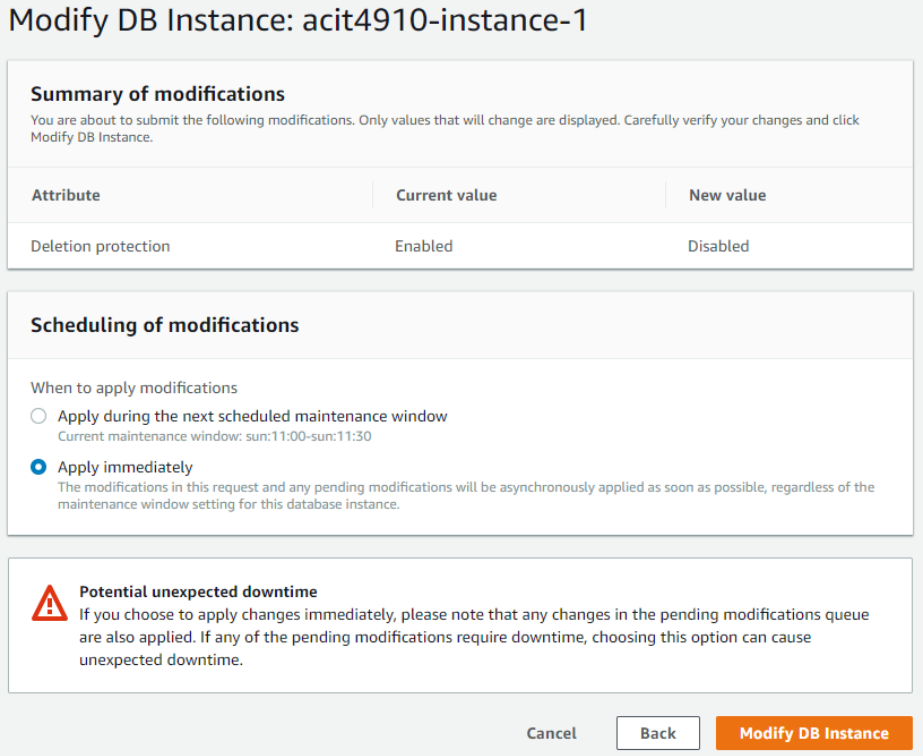


You need to disable the delete protection before you can delete your instance.

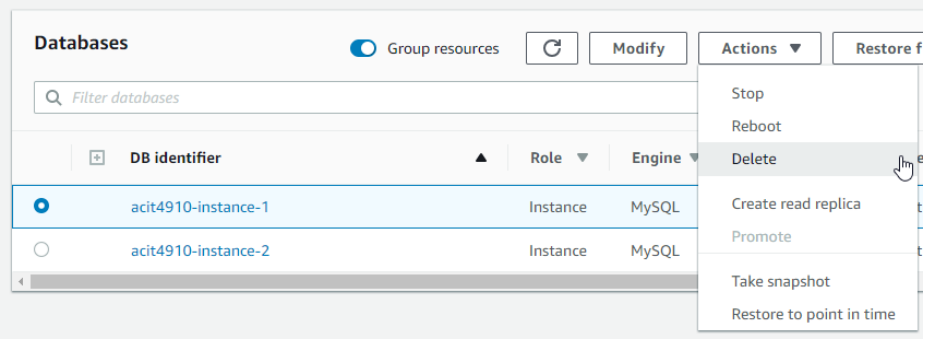


Click Continue.

Click "Apply Immediately" and click "Modify DB Instance".



The "Delete" Action will now be available from the "Actions" Dropdown.

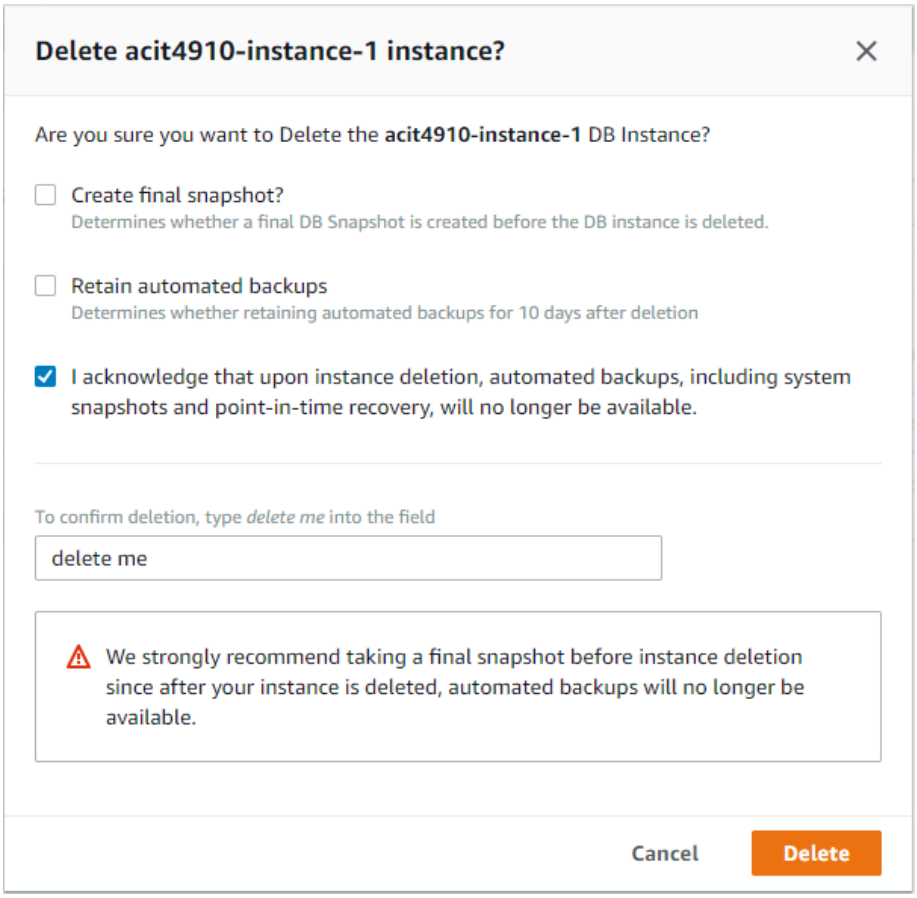


In our case we don't want to keep *any* snapshots so unselect create a final snapshot, and unselect retain automated backups.

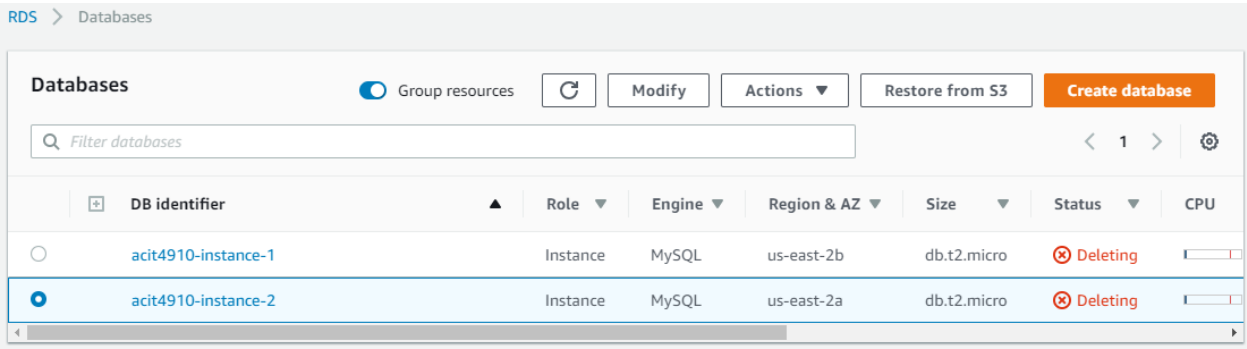
Click that you acknowledge that upon deletion your database is **GONE**!

To confirm type: *delete me* in the text field.

Click Delete.



Repeat for `comp3920-instance-2`.



Done.